BRAINSTEM

Digital Reviewer/Notes for STEM Subjects

First Term

CHEMISTRY IN THE FIELD OF HEALTH

Physical and Chemical Properties of Matter

Physical Change
- alters the appearance, size, or shape without forming new identity, often reversible.

Chemical Change
- produces one or more new substances with unique properties, usually difficult to reverse

Observation without change
- properties like color, state of matter, and odor can be noted without altering the substance's chemical makeup.

Measurable Constants
- density, mass, volume, melting point, and boiling point provide quantitative data for identification.

Conductivity and State
- the ability to conduct heat or electricity and the current state (solid, liquid, gas) are key physical markers.

Chemical Properties of Matter

Identity Transformation
- chemical properties describe a substance's ability to change into a different substance with new properties

OBSERVATION VIA CHANGE
- Unlike Physical Properties, chemical properties can only be observed when a substance is undergoing a chemical reaction

Irreversibility
- changes resulting from chemical properties are often difficult or impossible to reverse by physical means.
- Flammability (burning)
- Oxidation
- Reactivity with acids
- Toxicity and Corrosion

COMPARATIVE ANALYSIS

Observation Method
- P - without changing identity
- C - seen during a change

Outcome of Testing
- P - no new substance
- C - entirely news substances

Identity Retention
- P - as is
- C - potential to become

What Are Periodic Trends
- are predictable patterns in the properties of elements as you move across periods and down groups
- these trends help scientists predict: atomic size, ease of removing electrons, attraction for electrons, chemical behavior

Main Trends
- atomic radius
- ionization energy
- electron affinity
- electronegativity

Trend 1: Atomic Radius
- the distance from the nucleus to the outermost electron
- across a period
- atomic radius decreases

Reason:
- more protons - stronger nuclear attraction - electrons pulled together
- Down a group
- atomic increasing

Trend 2: Ionization Energy
- the energy required to remove one electron from a gaseous atom
- across a period
- ionization energy increases
- atoms become smaller
- electrons are held more tightly
- down a group
- ionization energy decreases
- **REASON:** outer electrons are farther from the nucleus
- Left to right - increases
- Top to bottom - decreases

Trend 3: Electron Affinity
- the energy change when an atom gains an electron
- higher (more negative)
  -atoms like gaining electrons
- across a period
  -generally increases
- down a group
  -generally decreases
- important note
  -noble gases usually have little or no affinity

Trend 4: Electronegativity
- the ability of an atom to attract bonding elements
- across a period
  -increases
- down a group
  -decreases

Summary

Property	Across Period	Down Group
Atomic radius	⬇	⬆
Ionization	⬆	⬇
Energy		⬇
Electron Affinity	⬆	⬇
Electronegativity	⬆	⬇

CHEMISTRY IN THE FIELD OF HEALTH

Naming Compounds

1. Ionic Compounds

- metal + non-metal
- rule: name of the cation (metal) first, then the anion (non-metal)
- for the non-metal, change the ending to -ide

Formula	IUPAC Name
NaCl	Sodium chloride
MgO	Magnesium oxide
CaBr2	Calcium bromide
Al2O3	Aluminum oxide
K2S	Potassium sulfide

Metals with variable charges

- use Roman numeral to indicate the charge of the metal
- examples:
- FeCl2 - iron (III) chloride
- Cu2O - copper (I)oxide
- CuO - copper (II) oxide
- FeCl3 - iron (III) chloride

2. Ionic Compounds with Polyatomic Ions

- is a group of atoms carrying an overall charge element name + polyatomic ion
- Rule: name the cation first, followed by the polyatomic ion
- examples:
- NaOH - Sodium hydroxide
- CaCo - Calcium Carbonate
- KNO3 - potassium nitrate
- MgSO4 -magnesium sulfate

3. Covalent and Molecular Compounds

- non-metal + non-metal
- use prefixes to show the nuber of atoms

1. mono
2. di
3. tri
4. tetra
5. penta
6. hexa
7. hepta
8. octa
9. nona
10. deca

- Rules: name the first element
- use prefix, second element (-ide)
- mono- usually omitted for the first element
- examples:
- XO- Carbon monoxide
- CO2- Carbon dioxide
- N2O- Dinitrogen trioxide

THE POLARITY OF A MOLECULE BASED ON ITS STRUCTURE

POLARITY

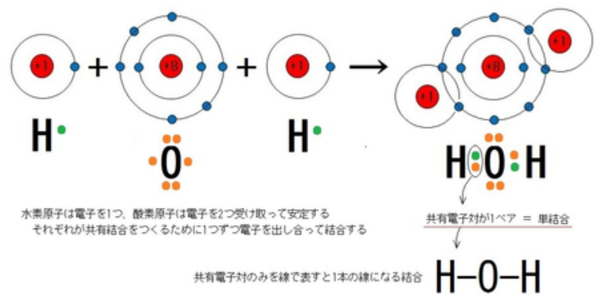
- equal or unequal sharing of electrons among the atoms of a molecule
- a molecule could be a group of atoms. It's tiniest unit that may participate during a chemical reaction. There are many different types of molecules, and each one of those molecules may be categorized into polar and non-polar groups.

POLAR MOLECULE

- there is unequal or asymmetrical distribution of electrons among the atoms of a molecule

OCTET RULE

- states that atoms tend to gain, share or transfer electrons in order to attain a stable 8 valence electron configuration.



NON-POLAR MOLECULE

- there is equal or symmetrical distribution of electrons among the atoms in a molecule

ELECTRONEGATIVITY

- relative ability of an atom to draw electrons in a bond toward itself.

BOND RANGE

Electronegativity Difference	Bond Type
0 - 0.4	Nonpolar covalent bond
0.5 - 2.0	Polar covalent bond
2.1 above (between metals and non-metals)	Ionic

CHEMISTRY IN THE FIELD OF HEALTH

POLARITY OF MOLECULES

(Molecular Geometry)

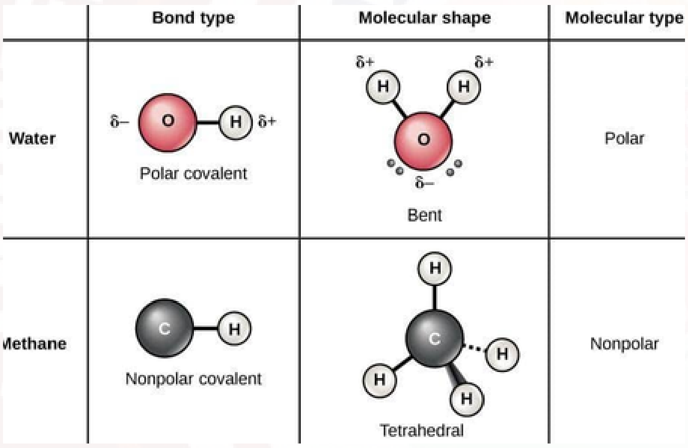
- pertains to the three-dimensional arrangement of atoms in a molecule

LEWIS ELECTRON DOT STRUCTURE (LEDS)

- valence electrons are represented using a dot around the symbol of the element

NONPOLAR MOLECULE

the shape of the molecule is symmetrical
- equal sharing of electrons
- no dipole moment
- valence electrons are shared equallt on both sides of an atom



POLAR	NON-POLAR
Polar Molecules have dipoles. Dipole moment is used to measure the polarity of a chemical bond between two atoms in a molecule	Non-polar molecules do not have a positive or negative ends
In polar molecules, we see that the charge is not uniformly distributed	Their dipole moments add up to zero (they cancel out)
Their dipole moments of polar molecules don't add up to zero (or don't cancel out)	It is electrically symmetric, that isthe electrical charges are uniformly distributed

POLAR	NON-POLAR
When highly electronegative atom bond with a comparatively less electronegative atom, a polar molecule is made	Most of the hydrocarbons liquids are non-polar. Nonpolar molecules do not interact the same way.
Polar molecules have electronegativity difference between 0.5 & 1.9	If two combining atoms have similar or equal elecronegativity values, the bond formed is non-polar.
A polar covalent bond is which one atom has a greater attraction for the electrons than the other atom. If this relative attraction is great enough, then the bond is an ionic bond.	Nonpolar molecules have electronegativity difference of 0.4 less. A nonpolar covalent bond is one in which the electrons are shared equally between two atoms.

Extensive Vs. Intensive Properties of Matter

Extensive properties
- depend on the amount of matter present
- properties that change when the amount of the matter changes

Intensive Properties
- do not depend on the amount of matter present
- properties that remain the same regardless of the amount of matter

Density = mass/volume

Why are intensive properties important
- identify substances
- maintain product quality
- manufacture medicines
- test fuel quality
- monitor environmental samples

CHEMISTRY IN THE FIELD OF HEALTH

WHAT IS ACCURACY?

- Refers to how close a measured value is to the true value or accepted value

PRECISION

- how close are repeated measurements to one another
- high precision means results are consistent

EXPERIMENTAL ERRORS

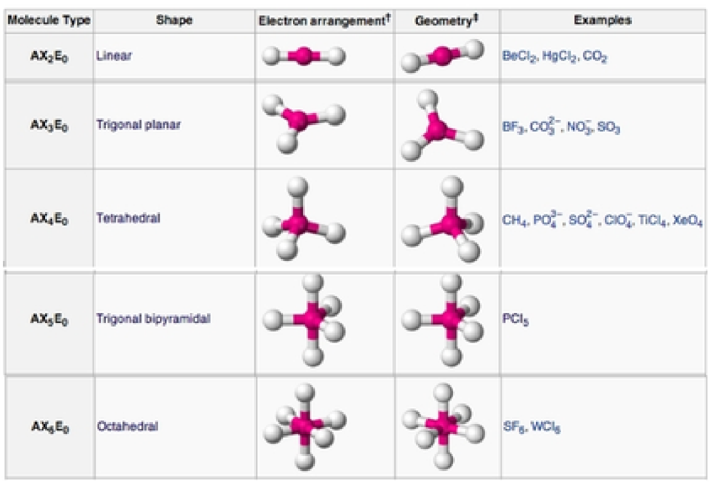
SYSTEMATIC ERROR

- occur in the same direction repeatedly

RANDOM ERROR

- scattered

ACCURACY	PRECISION
Close to true value	Close to each other
Concerned with correctness	Concerned with consistensy
Affected by systematic error	Affected bt random error



Molecule Type	Shape	Electron arrangement†	Geometry‡	Examples
AX_2E_0	Linear			$BeCl_2$, $HgCl_2$, CO_2
AX_3E_0	Trigonal planar			BF_3, CO_3^{2-}, NO_3^-, SO_3
AX_4E_0	Tetrahedral			CH_4, PO_4^{3-}, SO_4^{2-}, ClO_4^-, $TiCl_4$, XeO_4
AX_5E_0	Trigonal bipyramidal			PCl_5
AX_6E_0	Octahedral			SF_6, WCl_6